

HDOC - Day 7 Activity

C Programming Assignment

Question 1:

Write a C program to find final velocity when initial velocity, acceleration, and time are given.

Input, Output and Formula

Inputs: Initial Velocity (**u**) , Acceleration (**a**) & Time (**t**)

Output: Final Velocity (**v**)

Formula: $v = u + at$

Standard Test Case Table

Test case	Input(s)	Expected output	Actual output	Result
1	u=0, a=5, t=10	v = 50.00	v = 50.00	Pass
2	u=10, a=2, t=5	v = 20.00	v = 20.00	Pass
3	u=15, a=-3, t=4	v = 3.00	v = 3.00	Pass

Algorithm

1. Start.
2. Read initial velocity, acceleration, and time from the user.
3. Calculate final velocity using $v = u + at$.
4. Display the final velocity(v).
5. Stop.

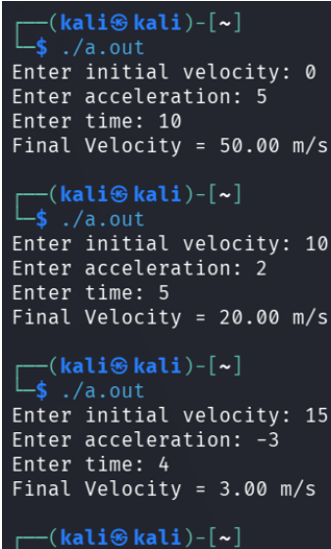
C Program

```
#include <stdio.h>

int main()
{
    float u, a, t, v;
    printf("Enter initial velocity: ");
    scanf("%f", &u);
    printf("Enter acceleration: ");
    scanf("%f", &a);
    printf("Enter time: ");
    scanf("%f", &t);

    v = u + (a * t);
    printf("Final Velocity = %.2f m/s\n", v);
    return 0;
}
```

Output :



```
(kali㉿kali)-[~]
└─$ ./a.out
Enter initial velocity: 0
Enter acceleration: 5
Enter time: 10
Final Velocity = 50.00 m/s

(kali㉿kali)-[~]
└─$ ./a.out
Enter initial velocity: 10
Enter acceleration: 2
Enter time: 5
Final Velocity = 20.00 m/s

(kali㉿kali)-[~]
└─$ ./a.out
Enter initial velocity: 15
Enter acceleration: -3
Enter time: 4
Final Velocity = 3.00 m/s

(kali㉿kali)-[~]
```

Question 2:

Write a C program to find potential energy when mass, acceleration due to gravity, and height are given.

Input, Output and Formula

Inputs: Mass (m) , Acceleration due to gravity (g) & Height (h)

Output: Potential Energy (PE)

Formula: $P = mgh$

Standard Test Case Table

Test case	Input(m,g,h)	Expected output(PE)	Actual output(PE)	Result
1	10 ,9.8 ,5	490 J	490 J	Pass
2	20 ,9.8 ,2	392 J	392 J	Pass
3	5 ,9.8 ,10	490 J	490 J	Pass

Algorithm

1. Start.
2. Read mass, gravity, and height from the user.
3. Calculate potential energy using $PE = mgh$.
4. Display the result.
5. Stop.

C Program

```
#include <stdio.h>
int main() {
float m, g, h, pe;
printf("Enter mass: ");
scanf("%f", &m);
printf("Enter gravity: ");
scanf("%f", &g);
printf("Enter height: ");
scanf("%f", &h);
pe = m * g * h;
```

```
printf("Potential Energy = %.2f Joules\n", pe);
return 0;
}
```

Output :

```
(kali㉿kali)-[~]
└─$ vi pe.c

(kali㉿kali)-[~]
└─$ cc pe.c

(kali㉿kali)-[~]
└─$ ./a.out
Enter mass: 10
Enter gravity: 9.8
Enter height: 5
Potential Energy = 490.00 Joules

(kali㉿kali)-[~]
└─$ ./a.out
Enter mass: 20
Enter gravity: 9.8
Enter height: 2 : by the name "potential.c.swp"
Potential Energy = 392.00 Joules Mon Aug 31 08:11:13 2020
File name: kali/potential.c

(kali㉿kali)-[~] YES
└─$ ./a.out File name: kali -- host name: kali
Enter mass: 5 : ID: 43345 (STILL RUNNING)
Enter gravity: 9.8 "potential.c"
Enter height: 10 FOUND
Potential Energy = 490.00 Joules g the same file. If this is the case,
... different instances of the same
```

Question 3:

Write a C program to find resistivity when resistance, area, and length are given.

Input, Output and Formula

Inputs: Resistance (R) ,Area (A) & Length (L)

Output: Resistivity (ρ)

Formula: $\rho = (R \cdot A) / L$

Standard Test Case Table

Test case	Input(R,A,L)	Expected output (ρ)	Actual output (ρ)	Result
1	10 ,2 ,5	4.00 $\Omega \cdot m$	4.00 $\Omega \cdot m$	Pass
2	15 ,3 ,5	9.00 $\Omega \cdot m$	9.00 $\Omega \cdot m$	Pass
3	20 ,4 ,10	8.00 $\Omega \cdot m$	8.00 $\Omega \cdot m$	Pass

Algorithm

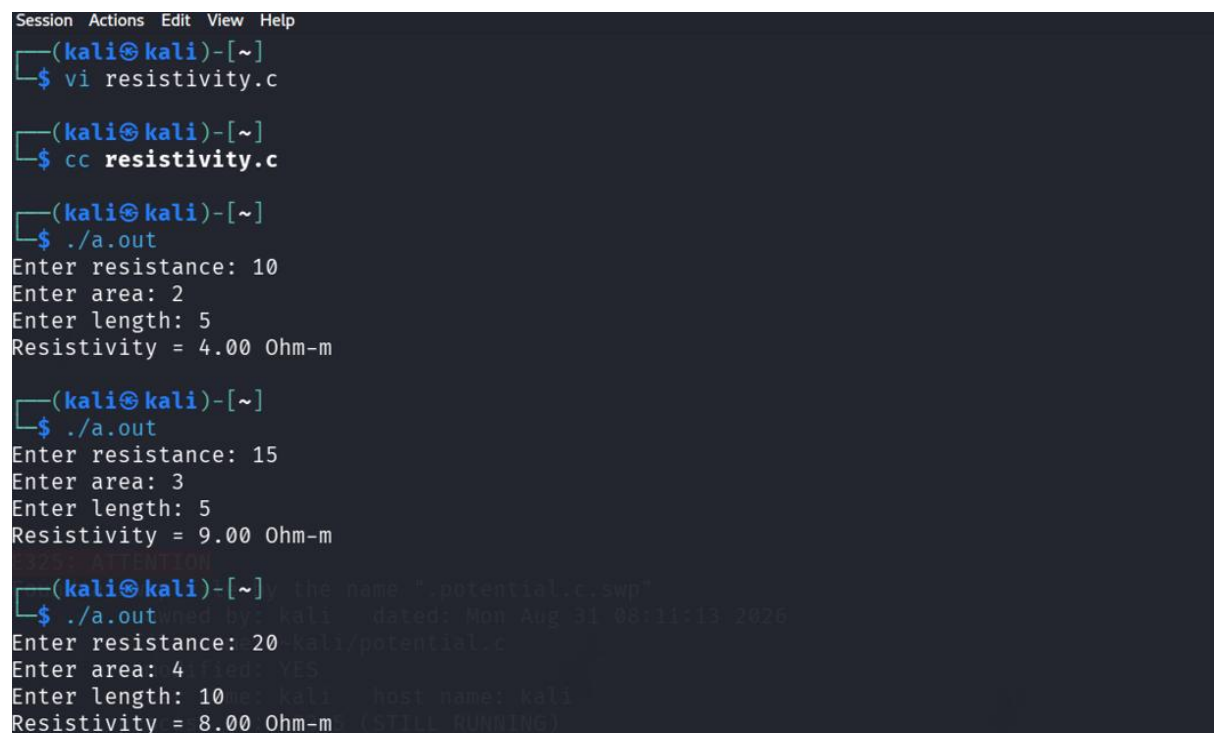
1. Start.
2. Read resistance, area, and length.
3. Calculate resistivity using $\rho = (R \times A) / L$.
4. Display the resistivity.
5. Stop.

C Program

```
#include <stdio.h>

int main() {
    float R, A, L, rho;
    printf("Enter resistance: ");
    scanf("%f", &R);
    printf("Enter area: ");
    scanf("%f", &A); printf("Enter length: ");
    scanf("%f", &L);
    rho = (R * A) / L;
    printf("Resistivity = %.2f Ohm-m\n", rho);
    return 0; }
```

Output :



```
Session Actions Edit View Help
(kali㉿kali)-[~]
$ vi resistivity.c

(kali㉿kali)-[~]
$ cc resistivity.c

(kali㉿kali)-[~]
$ ./a.out
Enter resistance: 10
Enter area: 2
Enter length: 5
Resistivity = 4.00 Ohm-m

(kali㉿kali)-[~]
$ ./a.out
Enter resistance: 15
Enter area: 3
Enter length: 5
Resistivity = 9.00 Ohm-m

(kali㉿kali)-[~]
$ ./a.out
Enter resistance: 20
Enter area: 4
Enter length: 10
Resistivity = 8.00 Ohm-m
```

NOTICE: the name ".potential.c.swp" used by: kali - dated: Mon Aug 31 08:11:13 2026
kali/potential.c
dated: YES
host: kali - host name: kali
(STILL RUNNING)

